

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Answer:

1) **Given outcomes:**

H H T H H H T H H T

- Total tosses (n) = 10
- Number of Heads (H) = 7

MLE Formula:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$\hat{p} = \frac{7}{10} = 0.7$$

2) **Given:**

- Total students = 50
- Passed = 42

MLE Formula:

$$\hat{p} = \frac{42}{50} = 0.84$$

3) **Given:**

- Total bulbs = 100
- Defective bulbs = 8

MLE Formula:

$$\hat{p} = \frac{8}{100} = 0.08$$

4) **Given:**

- Total rolls = 60
- Number of sixes = 18

MLE Formula:

$$\hat{p} = \frac{18}{60} = 0.30$$

5) **Given**

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Gradient = 4
- Learning Rate = 0.01

(i) Predicted Output

Formula:

$$y = w \times x$$

$$y = 2 \times 4 = 8$$

(ii) Error

Formula:

$$\begin{aligned}\text{Error} &= \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2\end{aligned}$$

(iii) Updated Weight

Formula:

$$w_{\text{new}} = w - (\eta \times \text{Gradient})$$

where

- $w = 2$
- $\eta = 0.01$
- $\text{Gradient} = 4$

$$\begin{aligned}w_{\text{new}} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96\end{aligned}$$